UInt64 same-support classifier correspondence

Producer proof packet; independent review pending

30 October 2026

Checked statement

For a prime modulus $5 \le p < 2^{31}$ and canonical UInt64 inputs $s_1, t_1, s_2, t_2 < p$, the literal word-value classifier equals the existing field classifier after decoding into $\mathbf{Z}/p\mathbf{Z}$. The order is inverse first, equal second, zero subtraction third, otherwise mixed. The theorem stops before doubling or other payload computations.

Connection proved

Word negation is $(p - (b \bmod p)) \bmod p$. Since $b \bmod p < p$, subtraction does not wrap. Word subtraction is $(a + \mathrm{neg}(b)) \bmod p$. Canonicality and $p < 2^{31}$ bound the sum below 2^{64}. Existing residue equality/negation/subtraction lemmas then translate every predicate while preserving branch order. The all-zero overlap is kernel-checked to select the inverse branch.

The new proof uses actual Lean UInt64 operations, not merely a natural-number formula. The connection from the Rust source to these expressions remains a manual fragment mapping; no Rust compiler-correctness theorem is asserted.

Evidence

NatClassifier and WordClassifier compiled successfully. A separate native process checked the final type and five named axiom readbacks; only propext, Classical.choice and Quot.sound occur. Compile commands were individually bounded at 120 seconds. Existing pinned Mathlib Field.ZMod objects supplied the missing field instance; five files were copied into a private dependency union. No installation, download, scientific trial or runtime cost comparison occurred.

Retained failures

The first NatClassifier attempt failed on dependent rewrite motives. Replacing that proof step with simplification preserved the statement. A word-subtraction attempt failed because simplification had changed the expression before its no-wrap inequality matched. The corrected proof supplies the inequality in the normalized expression. A prematurely launched API probe failed before its imported object existed. All native receipts remain. Earlier missing-field-instance attempts remain in the original packet.

Source and remaining obligations

The Rust fragments are Field::neg/sub in three_leads.rs and the inverse/equal/zero tests at projective_law_gate.rs:921–933. The $n = 1$, table-free, odd-prime route is intended. Counters, loop-state side effects, and exceptional execution are outside this value theorem. Prior multiplication lemmas can support the next composition of 16 multiplies and four squares; that composition and reachable-input validity are not asserted here. The field classifier itself is a manually defined abstraction. Separate source-fidelity review remains required before promotion.